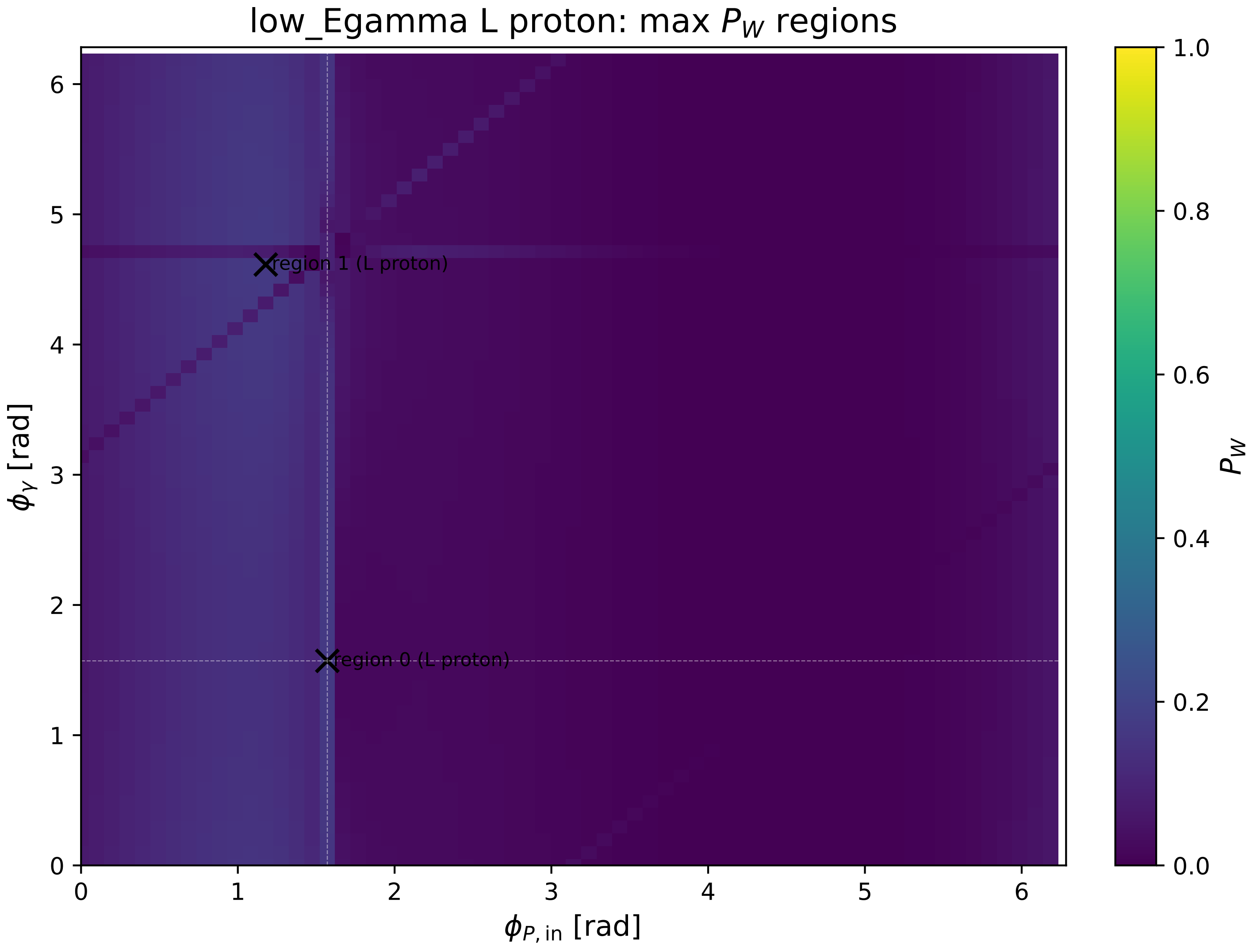
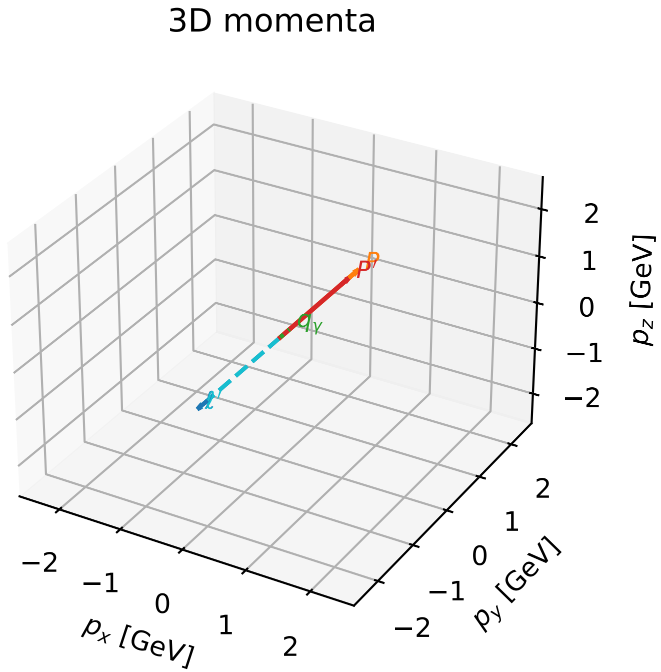


low_Egamma L proton: max P_W regions
region 1 (L proton)
region 0 (L proton)
ϕ_γ [rad]
$\phi_{P,\mathrm{in}}$ [rad]
P_W
0
1
2
3
4
5
6
0.0
0.2
0.4
0.6
0.8
1.0

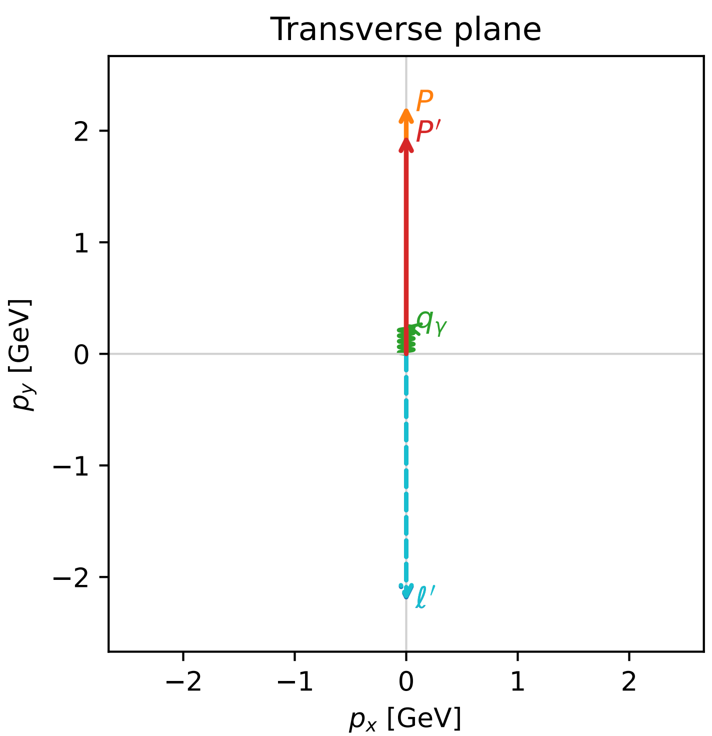
L proton characteristic max P_W configuration

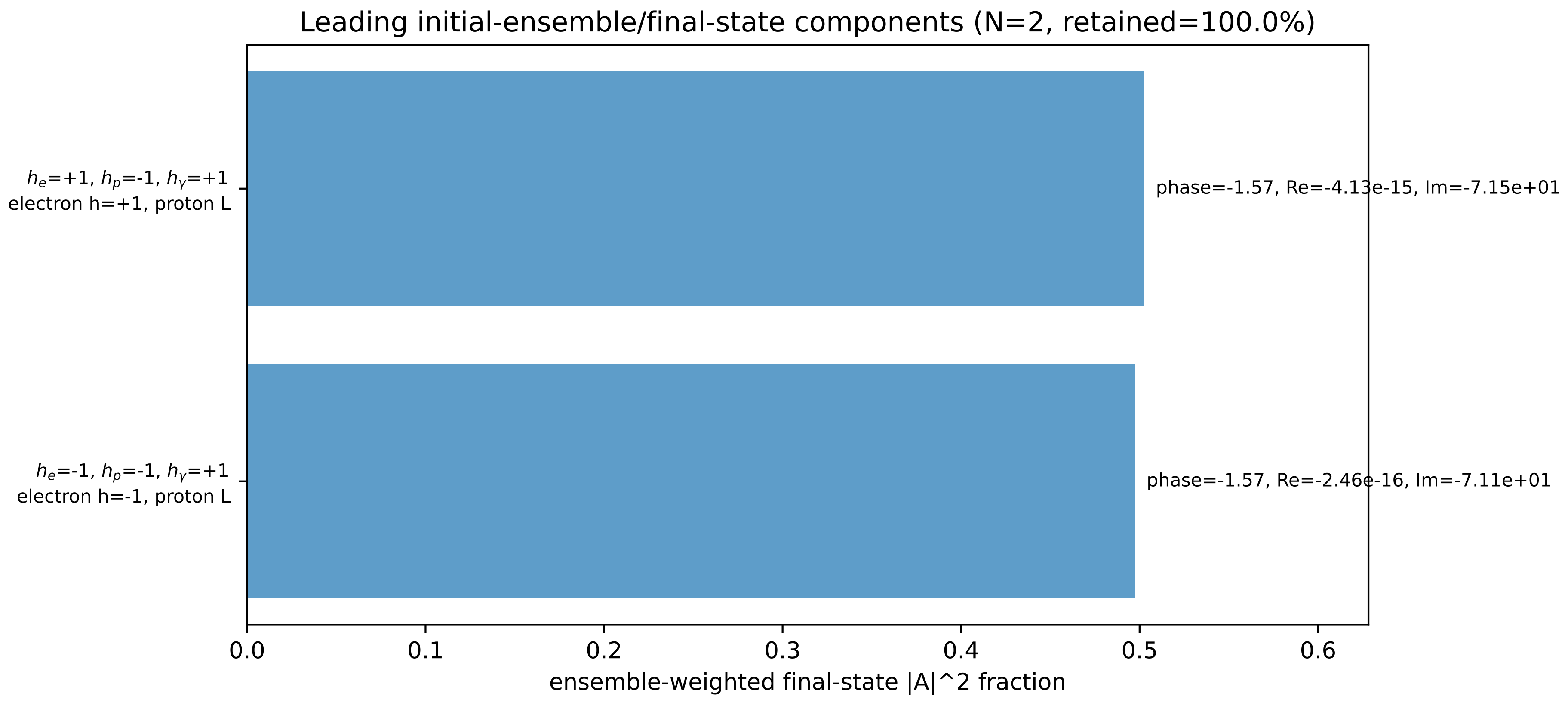


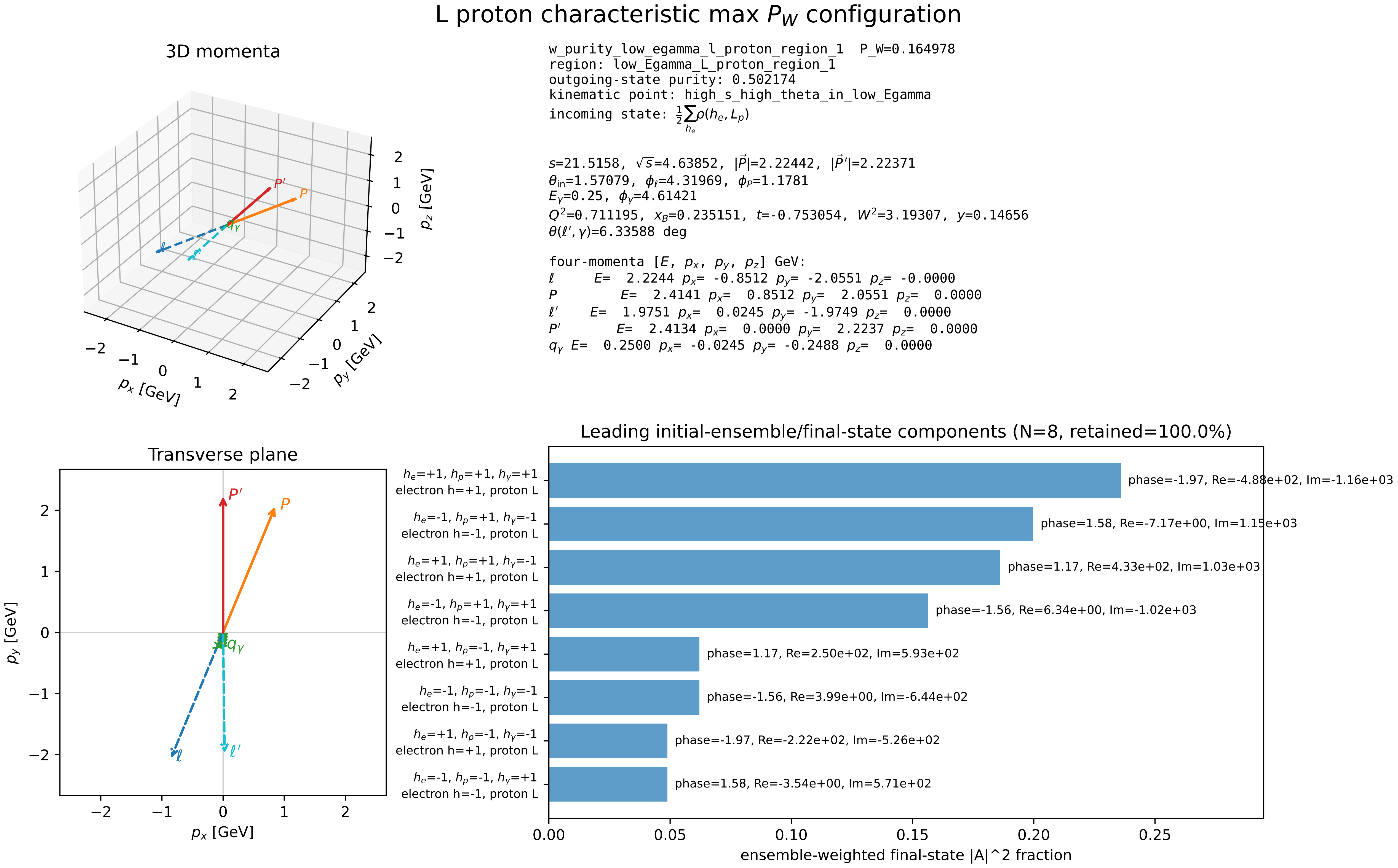
w_purity_low_egamma_l_proton_region_0 P_W=0.165806
region: low_Egamma_L_proton_region_0
outgoing-state purity: 0.500013
kinematic point: high_s_high_theta_in_low_Egamma
incoming state: $\frac{1}{2}\sum_{h_e}\rho(h_e, L_p)$

s=21.5158, $\sqrt{s}$=4.63852, $|\vec{P}|$=2.22442, $|\vec{P}'|$=1.96296
θ_{in}=1.57079, ϕ_ℓ=4.71239, ϕ_P=1.5708
E_γ=0.25, ϕ_γ=1.5708
Q^2=1.56301e-11, x_B=1.47039e-10, t=-0.0114583, W^2=0.986143, y=0.00515116
$\theta(\ell', \gamma)$=180 deg

four-momenta [E, p_x, p_y, p_z] GeV:
ℓ E= 2.2244 p_x= -0.0000 p_y= -2.2244 p_z= -0.0000
P E= 2.4141 p_x= 0.0000 p_y= 2.2244 p_z= 0.0000
ℓ' E= 2.2130 p_x= -0.0000 p_y= -2.2130 p_z= 0.0000
P' E= 2.1756 p_x= 0.0000 p_y= 1.9630 p_z= 0.0000
q_γ E= 0.2500 p_x= 0.0000 p_y= 0.2500 p_z= 0.0000






L proton characteristic max P_W configuration
3D momenta
p_x [GeV]
p_y [GeV]
p_z [GeV]
P'
P
q_γ
ℓ
ℓ'
w_purity_low_egamma_l_proton_region_1 P_W=0.164978
region: low_Egamma_L_proton_region_1
outgoing-state purity: 0.502174
kinematic point: high_s_high_theta_in_low_Egamma
incoming state: $\frac{1}{2}\sum_{h_e}\rho(h_e, L_p)$
s=21.5158, $\sqrt{s}$=4.63852, $|\vec{P}|$=2.22442, $|\vec{P}'|$=2.22371
θ_{in}=1.57079, ϕ_ℓ=4.31969, ϕ_P=1.1781
E_γ=0.25, ϕ_γ=4.61421
Q^2=0.711195, x_B=0.235151, t=-0.753054, W^2=3.19307, y=0.14656
$\theta(\ell', \gamma)$=6.33588 deg
four-momenta [E, p_x, p_y, p_z] GeV:
ℓ E= 2.2244 p_x= -0.8512 p_y= -2.0551 p_z= -0.0000
P E= 2.4141 p_x= 0.8512 p_y= 2.0551 p_z= 0.0000
ℓ' E= 1.9751 p_x= 0.0245 p_y= -1.9749 p_z= 0.0000
P' E= 2.4134 p_x= 0.0000 p_y= 2.2237 p_z= 0.0000
q_γ E= 0.2500 p_x= -0.0245 p_y= -0.2488 p_z= 0.0000
Transverse plane
p_x [GeV]
p_y [GeV]
Leading initial-ensemble/final-state components (N=8, retained=100.0%)
h_e=+1, h_p=+1, h_γ=+1 electron h=+1, proton L — phase=-1.97, Re=-4.88e+02, Im=-1.16e+03
h_e=-1, h_p=+1, h_γ=-1 electron h=-1, proton L — phase=1.58, Re=-7.17e+00, Im=1.15e+03
h_e=+1, h_p=+1, h_γ=-1 electron h=+1, proton L — phase=1.17, Re=4.33e+02, Im=1.03e+03
h_e=-1, h_p=+1, h_γ=+1 electron h=-1, proton L — phase=-1.56, Re=6.34e+00, Im=-1.02e+03
h_e=+1, h_p=-1, h_γ=+1 electron h=+1, proton L — phase=1.17, Re=2.50e+02, Im=5.93e+02
h_e=-1, h_p=-1, h_γ=-1 electron h=-1, proton L — phase=-1.56, Re=3.99e+00, Im=-6.44e+02
h_e=+1, h_p=-1, h_γ=-1 electron h=+1, proton L — phase=-1.97, Re=-2.22e+02, Im=-5.26e+02
h_e=-1, h_p=-1, h_γ=+1 electron h=-1, proton L — phase=1.58, Re=-3.54e+00, Im=5.71e+02
ensemble-weighted final-state |A|^2 fraction